इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY
सत्यमेव जयते

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NIELIT

Ultrasound and Beamforming

Starting from zero — no background assumed



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Before We Begin

This session assumes you know nothing about any of this. That is fine.

NO PHYSICS	we will build up from shouting at a wall
NO MATHS YET	every formula arrives only after the picture that explains it
ONE IDEA	each slide carries one idea, and we check it before moving on
ASK	if a word is unfamiliar, stop the session. Someone else is wondering too

By the end you will understand what a beamformer is and why it is hard to build.
That is all. The engineering detail comes in the sessions after this one.

The only thing to remember right now
ultrasound is an echo, and a beamformer is a way of listening to echoes carefully

Step 1 — You Already Know What an Echo Is

Shout at a wall. Count. That is the whole idea.

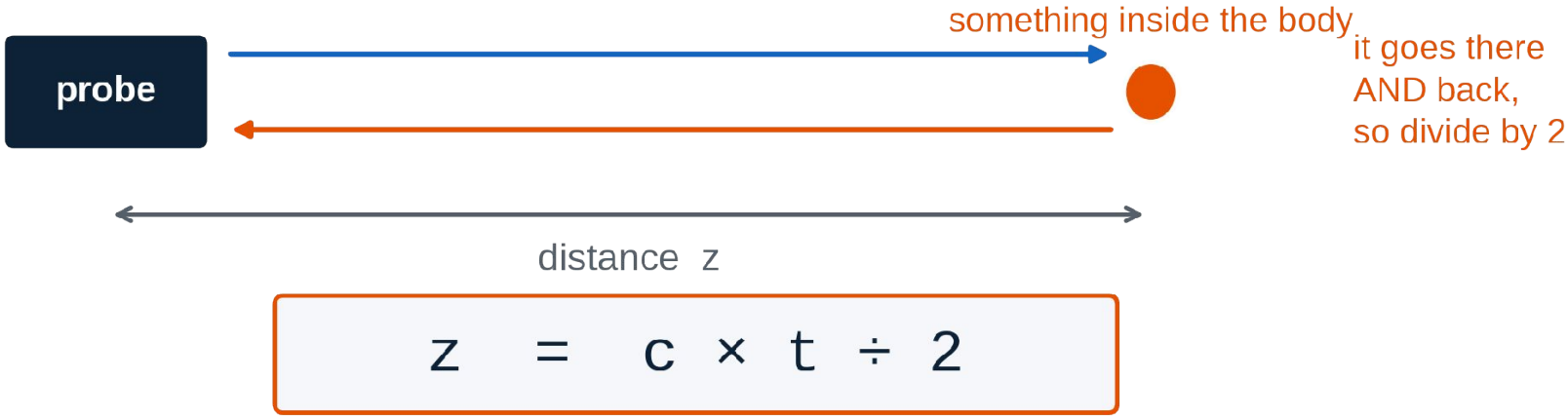


Count the seconds and you know how far away the wall is.

Sound travels at a known speed. If you know the time, you know the distance.

Step 2 — The Same Trick, Inside the Body

A probe sends a pulse and listens for the echo.



SYMBOL	WHAT IT MEANS
c	the speed of sound in tissue — about 1540 metres per second
t	the time between sending the pulse and hearing the echo

A Number That Will Keep Coming Back

Let us put real numbers into that formula, once.

AN ECHO COMES BACK AFTER 26 MICROSECONDS. HOW DEEP IS IT?

```
z  =  c × t ÷ 2
   =  1540 m/s  ×  26 µs  ÷  2
   =  0.020 m  =  20 mm
```

So time IS depth

Every microsecond of waiting is about 0.77 mm deeper into the body.

And it sets the frame rate

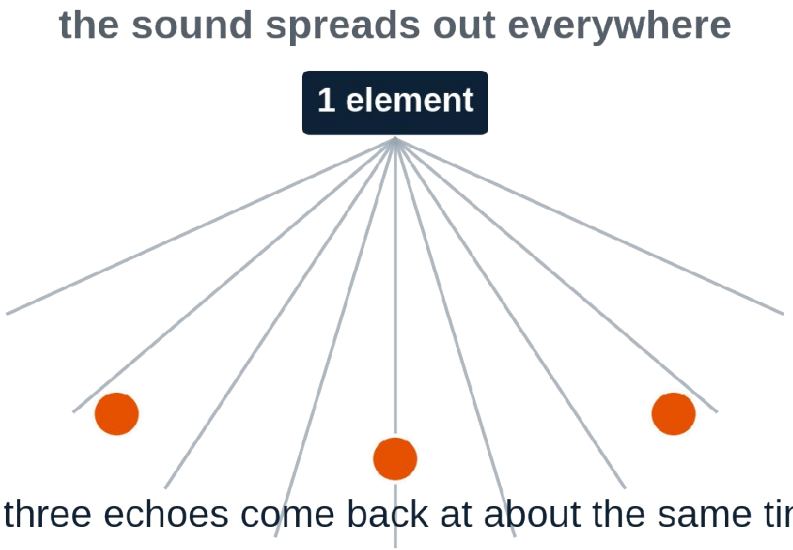
To see 20 cm deep you must wait 260 µs for the echo. Nothing can make that faster.

That one formula — depth equals speed times time divided by two — is the foundation of everything else.

If you follow only one line from this session, follow that one.

Step 3 — But One Listener Is Not Enough

It tells you how far. It cannot tell you which direction.

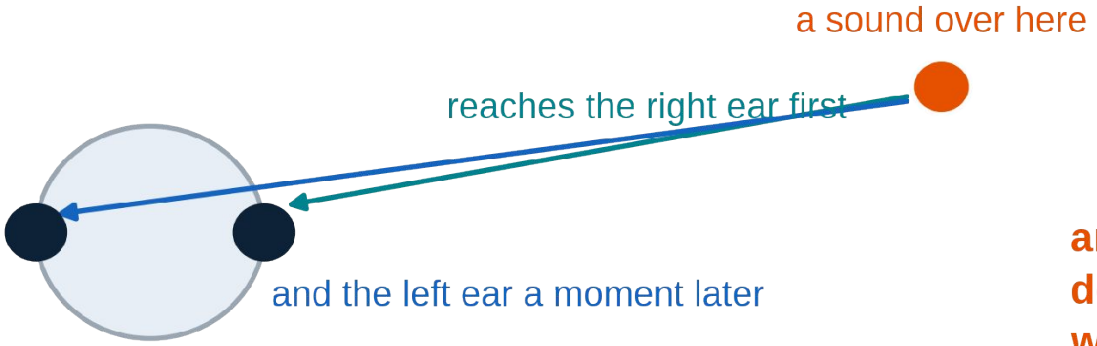


so you know HOW FAR, but not WHICH WAY

With one element you can measure depth, but you cannot build a picture.

Step 4 — How You Solve This Every Day

You have two ears, and you already do this without thinking.



your brain compares the two arrival times
and works out the direction

an ultrasound array does exactly this, with 64 ears

This is the key idea of the whole session
Direction is hidden in the difference between arrival times. More listeners means more differences to compare, and a better answer.

Step 5 — So We Use Many Listeners

A probe is not one element. It is a row of them — typically 64 to 256.

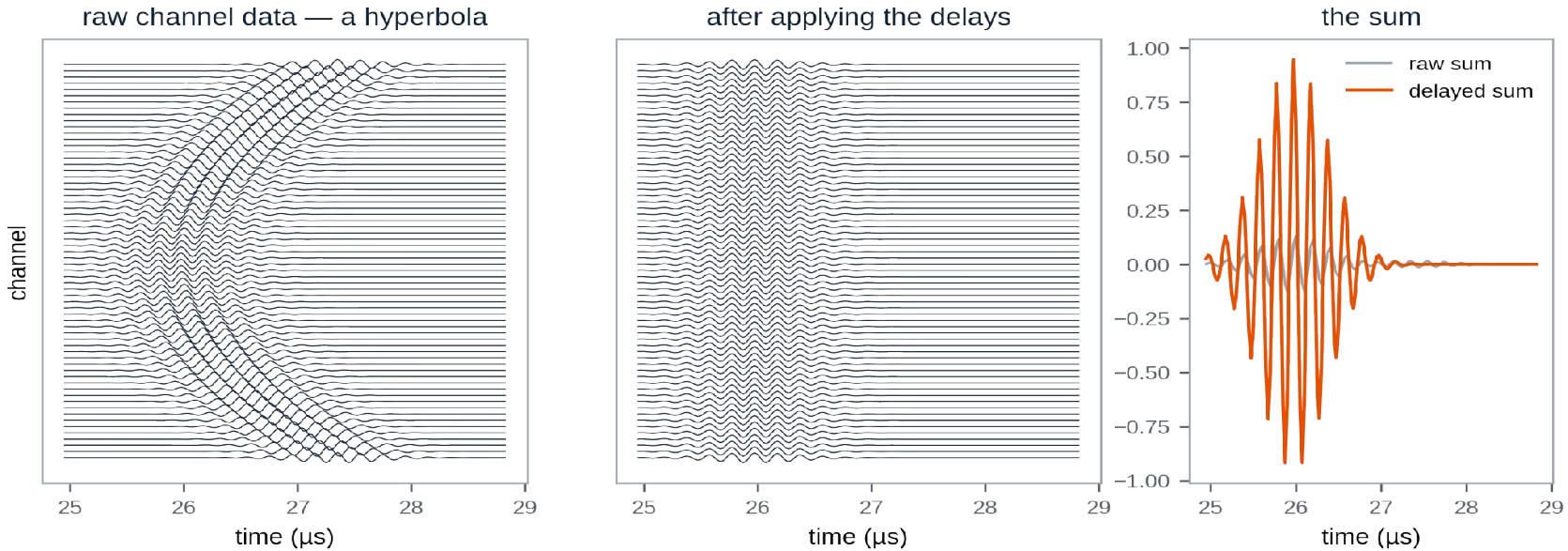
	ONE ELEMENT	AN ARRAY OF ELEMENTS
Tells you	how far away something is	how far AND in which direction
Picture	a single line of dots	a full two-dimensional image
Hardware	one wire	64 to 256 wires, each with its own amplifier and converter

Every element hears the same echo, but each one hears it at a slightly different moment.
Those small timing differences are the raw material a beamformer works with.

A word you will hear constantly
CHANNEL. One channel is one element plus all the electronics behind it — amplifier, converter, memory. When someone says “a 64-channel system” they mean 64 copies of all that, which is why these machines are expensive.

Step 6 — What the Elements Actually Hear

One small object, 64 elements, 64 slightly different recordings.



64 channels, point target at 20 mm. Aligning before summing raises the peak by 17.3 dB over summing them unaligned, and 36 dB over a single channel.

Left: what comes in. Middle: after shifting each one. Right: what you get if you add them.

Why the Curve?

Because the middle elements are closer to the object than the ends are.

THINK OF IT THIS WAY	SO THE PICTURE SHOWS
– the object is in front of the middle of the probe	– a curve, not a straight line
– so the middle elements are nearest to it	– deepest in the middle, rising at both ends
– their echo arrives first	– the shape is a hyperbola
– the elements at the ends are further away	– and its exact shape tells you where the object is
– their echo arrives a little later	– which is why we can undo it

The curve is not a problem to be removed. It is the information.
Undo the curve correctly and you have found the object. Undo it wrongly and you have found nothing.

Step 7 — Now the Whole Trick

Shift each recording back by the right amount, then add them together.



Same four signals. The only difference is whether you line them up first.

Add them as they are: they fight each other. Line them up first: they help each other.

That Is Beamforming

Two words. Delay, then sum.

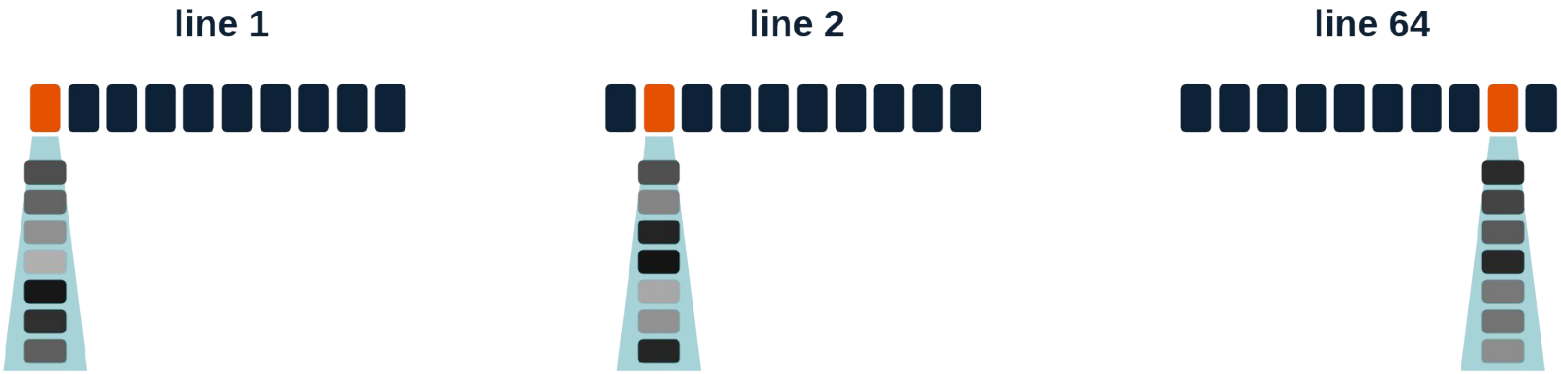
DELAY	shift each channel back by the time its echo was late, so every channel is describing the same instant
SUM	add them. The real echo adds up. Everything else — noise, echoes from elsewhere — does not line up, so it cancels

Delay-and-sum. You now know what the most widely used algorithm in medical ultrasound does.
Every other method in this course is a variation on these two steps.

Written down, it looks like this
B = sum over all channels of (weight × channel signal, shifted by its own delay)

“OK, But That Is One Point”

Correct. One delay-and-sum gives you the brightness of a single spot. Now we repeat it.



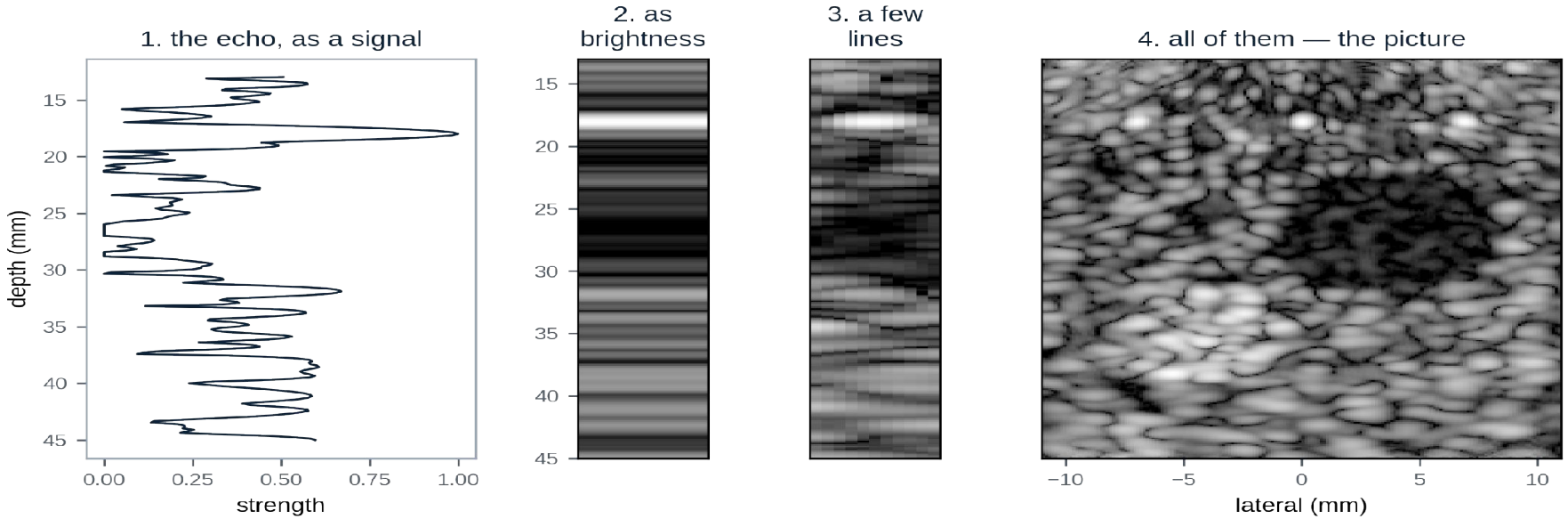
The probe listens along one line, paints it as a column of greys, then moves to the next line.

Sixty-four columns side by side make one frame. Twenty to fifty frames a second looks like live video.

So a picture is not taken, it is assembled
There is no camera and no lens. Each point in the image is a separate calculation, and the machine does a few million of them every second.

How an Echo Becomes a Picture

Four steps, left to right. This is the part that is hardest to imagine.



A strong echo is painted white, a weak one dark. Stack the lines side by side and the picture appears.

A strong echo is painted white, a weak echo dark. Nothing more clever than that.

What the Greys Actually Mean

Worth being precise, because it is easy to misread an ultrasound image.

WHAT YOU SEE	WHAT IT MEANS
`Bright white`	a strong echo — a big change in material, like tissue meeting bone
`Mid grey`	a weak scattered echo — ordinary soft tissue
`Black`	almost no echo came back — fluid, blood, a cyst
`Grainy texture`	not noise, and not real detail — it is speckle, explained below

Speckle, since it dominates every image

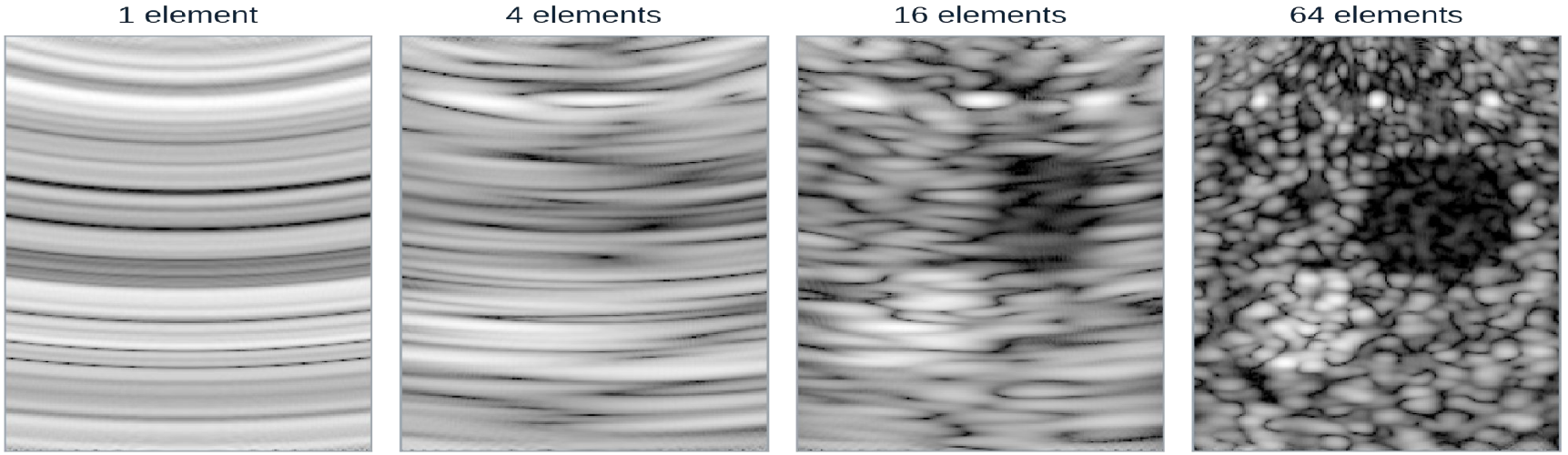
Tissue contains countless scatterers far smaller than the wavelength. Their echoes interfere, producing a grainy pattern. It looks like texture but it is not anatomy — two scans of the same tissue give different grain.

The image shows where echoes came from and how strong they were. It does not show what the tissue is.

A radiologist infers the tissue. The machine only ever reports echoes.

One Sensor or Many — What Actually Changes

The same echoes, beamformed with different numbers of elements.

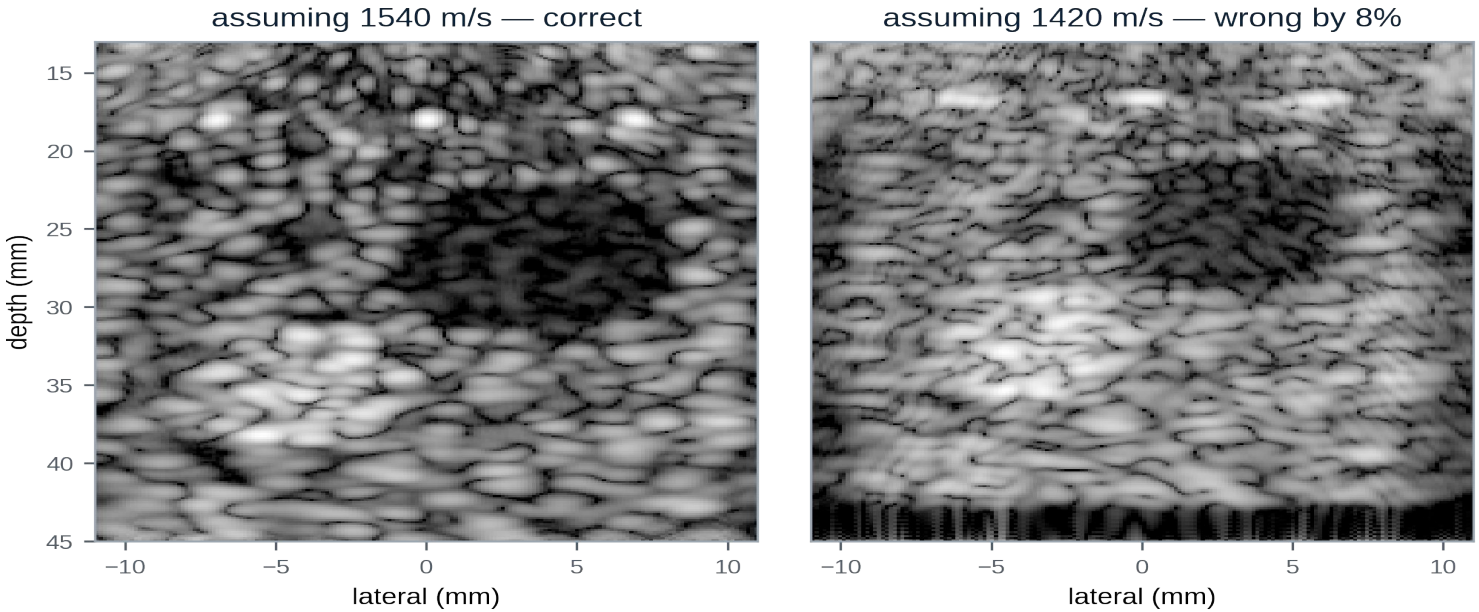


Exactly the same echoes every time. The only thing that changes is how many elements are listening and being combined.

With one element you get depth only, so everything smears into horizontal stripes. Detail is the reward for combining many.

“How True Is the Picture?”

The right question to ask, and the honest answer is: it is a careful guess.



Identical echoes. The machine simply guessed the speed of sound wrongly, and the picture fell apart.

Same echoes in both. The only difference is the speed of sound the machine assumed.

What the Machine Has to Assume

Every one of these is slightly untrue, and each has a visible consequence.

	THE ASSUMPTION	WHAT HAPPENS WHEN IT IS WRONG
Speed of sound	1540 m/s everywhere	fat is nearer 1450 — the image blurs and shifts
Straight paths	sound travels in straight lines	it bends at interfaces — things appear displaced
One bounce	each echo reflected once	it can bounce twice — ghost copies appear deeper
No obstruction	sound reaches everywhere	bone and gas block it — a shadow behind them

An ultrasound image is a reconstruction built on assumptions, not a photograph.
Learning to read it is partly learning which parts of it to distrust.

Why an engineer should care

No amount of arithmetic precision in your beamformer can fix a wrong speed of sound. That is a limit set by the patient, not by your hardware — and it is worth knowing which errors you can design away and which you cannot.

How Much Does It Help?

Worth knowing, because it decides how much hardware you buy.

more ears listening together = a clearer signal

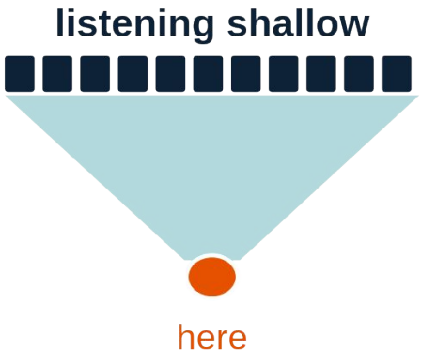


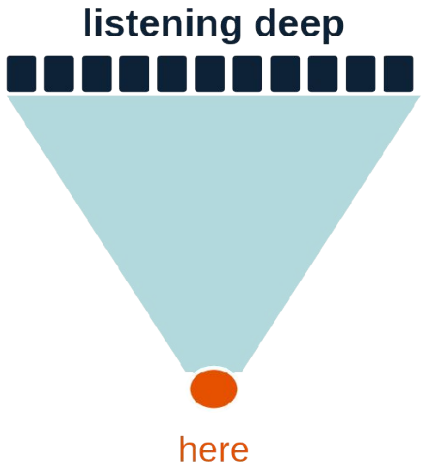
Every time you double the number of elements you gain 3 dB — and pay for twice the hardware.

Read that carefully
Going from 64 to 256 elements is four times the hardware for 6 dB. That is the uncomfortable arithmetic behind every ultrasound machine ever designed.

Step 8 — Choosing Where to Listen

Change the delays and the array listens somewhere else.





Change the delays and you change where the array is listening. Nothing moves.

Nothing moves. No lens, no motor. Only numbers change.

And We Can Change It Continuously

The echo from deep inside arrives later — so we refocus while we listen.

FIRST µs	echoes from shallow tissue arrive — focus the array shallow
LATER	echoes from deeper arrive — move the focus deeper
CONTINUOUSLY	keep doing this, for every depth, for all 64 channels, in real time

This is called dynamic receive focusing, and it is the reason ultrasound went digital.
You cannot change an analogue delay line thousands of times per image. You can change a number.

And it is the hard part
A new set of 64 delays, for every depth sample, for every line, for every frame. That is the computation this course is about building.

Step 9 — Turning Some Elements Down

A small refinement with a large effect on picture quality.

everyone shouts equally



loud, but messy at the edges

edges whisper



a little blurrier, much cleaner

Bar height = how much each element contributes. Turning the outer ones down is called apodisation.

What you lose
a slightly wider beam — so slightly blurrier

What you gain
far less interference from things you were not aiming at

Recap Before We Go On

Nine steps. Check you are comfortable with each.

1	an echo tells you distance, because sound has a known speed
2	depth = speed × time ÷ 2
3	one listener gives distance but not direction
4	two ears find direction by comparing arrival times
5	a probe uses 64 to 256 listeners, called channels
6	their recordings form a curve, and the curve holds the position
7	delay each one, then sum — that is beamforming
8	changing the delays changes where you are listening
9	turning the outer elements down cleans up the picture

If any of these is unclear, now is the moment to say so — everything after this builds on them.

Part 2 — Why This Is Hard to Build

The idea is simple. Doing it fast enough is not.

WHAT HAS TO HAPPEN	HOW OFTEN
`Work out a delay for one channel`	millions of times a second
`Multiply by a weight and add`	once per channel, per depth, per line
`Finish a whole image`	20 to 50 times a second
`All of it`	while running on a battery, inside a handheld probe

A 64-channel probe produces about 3.8 gigabytes of data every second, continuously.
That is roughly a full-length film, every second, and it must be processed as it arrives.

Why not just use a computer?
A general-purpose processor does one thing at a time, very fast. This problem needs thousands of small identical operations happening at once. That is what an FPGA or a custom chip is for — and that is why this is a VLSI course.

Counting in Whole Numbers

Hardware prefers integers. Here is why that matters.

	FIXED POINT	FLOATING POINT
What it is	whole numbers, with an agreed decimal place	numbers with a movable decimal point
Like	counting in millimetres	scientific notation
In hardware	small and fast	several times larger and slower
Used where	the whole receive datapath	rarely, and only after the data rate drops

Why it matters so much here

Whatever you choose gets built 64 times over, once per channel. A choice that costs a little extra per channel costs a great deal in total.

In plain words

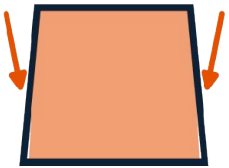
We measure everything in fixed steps, like using a ruler marked only in millimetres — and we choose the step size carefully.

One Trap Worth Knowing Now

When you add 64 numbers, where does the total go?

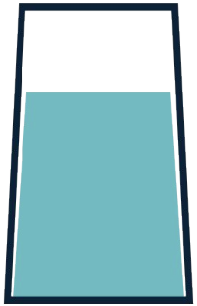
adding up 64 numbers needs somewhere big enough to put the total

a small bucket



it overflows
and you lose the answer

a big bucket



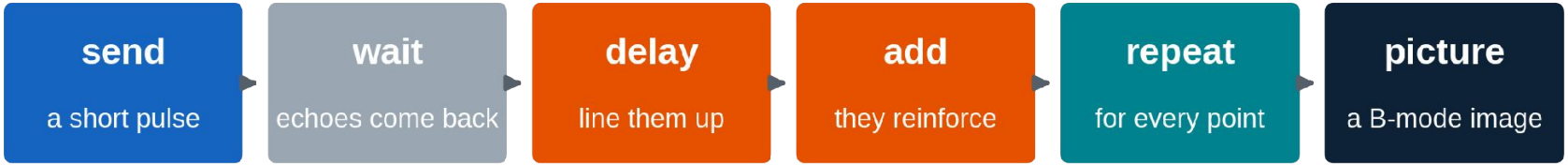
everything fits
nothing is lost

If the total does not fit, the answer is not just wrong — it is wrong in a way that looks like a real echo.

The Whole Thing, in One Picture

Everything you have seen, in order.

the whole of ultrasound imaging, in six steps



Steps 3 and 4 are the beamformer. That is the part this course is about.

You now know what the system does and which part of it this course builds.
The sessions that follow take steps 3 and 4 apart and put them into hardware.

Next
number formats, then the FPGA, then the real beamformer

Words You Now Know

Keep this slide. Every one of these will be used without explanation from now on.

WORD	WHAT IT MEANS
`Element`	one small piece of the probe that sends and listens
`Array`	the whole row of elements
`Channel`	one element plus its electronics
`Echo`	the sound that comes back from inside the body
`Delay`	how much you shift one channel before adding it
`Beamforming`	delaying all the channels and adding them
`Delay-and-sum`	the simplest and most common way of doing that
`Focus`	the place the array is currently listening to
`Apodisation`	turning the outer elements down to clean the picture
`Fixed point`	counting in whole steps, the way the hardware prefers
`Scan line`	one column of the image, built from one listening direction
`B-mode`	the ordinary 2-D grey picture, built from many scan lines
`Speckle`	the grainy texture — interference, not anatomy

Check Yourself

If you can answer these, you are ready for the next session.

1	An echo returns after 40 microseconds. How deep is the object?
2	Why can one element not produce a picture?
3	Why do the recordings from the different channels form a curve?
4	What are the two steps of delay-and-sum, and what does each one achieve?
5	You double the number of channels. What do you gain, and what do you pay?
6	Why must the place you store the total be bigger than the numbers you are adding?
7	How does a single number — one echo strength — turn into a picture on the screen?
8	Name one thing the machine has to assume, and what goes wrong when that assumption fails.

Answers: 31 mm · no direction · the middle elements are closer · delay then sum · 3 dB for twice the hardware · the sum exceeds any one term · brightness, repeated line by line · e.g. speed of sound, which blurs and shifts the image.

Thank You

Ultrasound and Beamforming — Starting from Zero

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